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The Hong Kong University of Science
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Clear Water Bay, Kowloon
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[Mei Zhou](#)

Education

Ph.D in Environmental Science
(May 2025)
Tübingen University, Germany
Score: Very good

M.S. in Energy Efficiency Engineering
(Jun 2020)
Xiamen University, China
GPA: 3.6/4.0

B.S. in Energy and Power Engineering
(Jun 2017)
Chongqing University, China
GPA: 3.5/4.0

Skills & Expertise

- Anaerobic and microaerobic microbe cultivation
- Analytical chemistry
- Shotgun metagenomic and 16S rRNA gene sequencing
- QIIME 2 for microbiome data analysis
- Experimental systems design, integration, operation, and maintenance

Curriculum Vitae

Mei Zhou

Postdoctoral researcher of Environmental Engineering at the Department of Civil and Environmental Engineering at the Hong Kong University of Science and Technology (HKUST). Centered on converting organic waste streams into the resources needed for a sustainable food, chemical, and energy economy, with emphasis on bioprocess engineering and the fundamental sciences of microbial communities that drive these processes.

Professional Experience

Aug 2025–Current

Postdoctoral Researcher

Department of Civil and Environmental Engineering
The Hong Kong University of Science and Technology,
Hong Kong
Research focus: Biogas biorefinery into value-added
bioproducts such as microbial protein and
polyhydroxybutyrate (PHB)
PI: Prof. Samir Kumar Khanal

Jul 2021–May 2025

Graduate Research Assistant

Department of Geoscience
Tübingen University, Germany
Research focus: Microaeration for valorization of hydrothermal
liquefaction process water
Supervisor: Prof. Largus T. Angenent and Dr. Joseph G. Usack

Sep 2017–Jun 2020

Graduate Research Assistant

College of Energy
Xiamen University, China
Research focus: Ammonia as an environmentally benign energy
and hydrogen carrier
Supervisor: Prof. Zhaolin Wang

Research Areas

- Energy and environment
- Microaeration and anaerobic digestion
- Gas fermentation
- Nutrient and resource recovery
- Nanobubble technology
- Photo-bio hybrid systems

Publications

1. Ntihuga, J. N., Usack, J. G., Mayer, R., Kinbokun, A. A., Yesil, H., **Zhou, M.**, & Angenent, L. T. (2026). Dried-bakery waste as a substrate for n-caproate and n-caprylate production via chain elongation. *Waste Management*, 216, 115454.
2. **Zhou, M.**, Taiwo, K., Wang, H., Ntihuga, J. N., Angenent, L. T., & Usack, J. G. (2024). Anaerobic digestion of process water from hydrothermal treatment processes: a review of inhibitors and detoxification approaches. *Bioresources and Bioprocessing*, 11(1), 47.
3. **Zhou, M.**, Usack, J. G., Smith, A. M., & Angenent, L. T. (2025). Toxicity and biodegradation of two different hydrothermal liquefaction process waters for anaerobic digestion and the effect of microaeration. *Chemical Engineering Journal*, 520, 165540.
4. **Zhou, M.**, Bello, I. O., Magdalena, J. A., & Usack, J. G. (2025). Development and biological validation of a flat-sheet membrane bioreactor for ultra-low gas flux bioprocesses. *Chemical Engineering Journal*, 166429.
5. Wang, W., **Zhou, M.**, Yang, H., & Shao, Z. (2024). Radiation resistance of graphene in tritiated water. *Fusion Engineering and Design*, 200, 114213.
6. **Zhou M.**, Chu Y.C., Wang Z.L., Chen J., & Zheng S. S. (2020). Emission characteristics of premixed NH₃-C₃H₈-air mixture for carbon reduction combustion. *Journal of Combustion Science and Technology*, 3, 1-8. (in Chinese)
7. **Zhou, M.**, Wang, Y., Chu, Y., Tang, Y., Tian, K., Zheng, S., ... & Wang, Z. (2019). Ammonia as an environmentally benign energy carrier for the fast growth of China. *Energy Procedia*, 158, 4986-4991.
8. Wang Y.G., **Zhou M.**, Wang Z.L., and Zheng S. S. (2018). Life-cycle energy efficiency estimation of large-scale ammonia fuel energy storage system. *Energy Storage Science and Technology*, 7(2), 301-308. (in Chinese)

Awards and Honors

1. Scholarship of China Scholarship Council (CSC)-Tübingen University joint Ph.D Program – Tübingen University – (Jul 2021–Jun 2025)
2. First Honor Graduate Studentship – Xiamen University – (Sep 2017–Jun 2020)
3. Second Prize in the 6th China Graduate Energy Equipment Innovation Design Competition Xiamen University Selection Competition – (2029)
4. Outstanding Graduate Cadre – Xiamen University – (2018)
5. First Prize in the 4th Energy Technology Innovation Competition – Xiamen University – (2018)

Conference Presentations (* denotes presenter)

1. **Zhou M.***, Khanal S.K., and Angenent L. T. When does microaeration enhance the valorization and stabilization of hydrothermal liquefaction (HTL) process water? 19th IWA World Conference on Anaerobic Digestion. Fueling sustainability, Jun 9–13, 2026, Valencia, Spain. Oral presentation.
2. **Zhou M.**, Ntihuga J. N.*, Angenent L. T., and Usack J. G. A new diffusion-driven flat-sheet membrane bioreactor for precise oxygen dosing in microaeration research. 18th IWA World Conference on Anaerobic Digestion. Bridging waste to value through anaerobic digestion, Jun 2–6, 2024, Istanbul, Turkey. Poster presentation.
3. **Zhou M.***, Wang Y.G., Chu Y.C., Tang Y. N., Tian K. K., Zheng S. S., Chen J., Wang Z.L. Ammonia as an environmentally benign energy carrier for the fast growth of China. 10th International Conference on Applied Energy, Aug 22–25, 2018, Hong Kong, China. Oral presentation.
4. **Zhou M.***, Wang Y.G., Wang Z.L., Chen J., Zheng S. S. Ammonia Progressing in China. 2018 AIChE Annual Meeting, Oct 28–Nov 02, 2018, Pennsylvania, USA. Oral presentation.

Certifications

1. Lab Safety Training on Chemical, Biological and Fire Safety, Jan 2026
2. Responsible Conduct of Research (1-RCR), The Hong Kong University of Science and Technology, Nov 2025

Relevant Courses

Graduate Courses

(M.S. in Energy Efficiency Engineering)

- Energy Chemistry and Chemical Engineering
- System Modeling and Simulation
- Energy Technology and Engineering
- Energy Harvesting Technology
- Structure and Principle of Energy and Power Systems
- Nature Dielectrics
- Photoelectric Measurement Techniques

Undergraduate Main Courses

(B.S. in Energy and Power Engineering)

- Engineering Thermodynamics
- Engineering Heat Transfer
- Engineering Fluid Mechanics
- Linear Algebra
- Advanced Mathematics
- Heat Exchanger
- Principles of Refrigeration&Cryogenics
- Principles of Steam Turbine
- Auto Control for Thermal Process
- Electrical and Electronics
- Foundation of Computer Information Management
- Foundation of Computer hardware technology
- Auto Control for Thermal Process
- Environmental Protection in Energy and Power Process
- Physics and New Technology